

OOPS ASSIGNMENT 9

Q1. Write a program to write numbers from 1 to 200 into a file.

Sol1:

```
#include <iostream>
```

```
#include <fstream> using
```

```
namespace std;
```

```
int main() {
```

```
    ofstream fout("NUM.TXT");
```

```
    for(int i = 1; i <= 200; i++) { fout
```

```
        << i << "\n";
```

```
    }
```

```
    fout.close();
```

```
    cout << "Numbers writtensuccessfully!";
```

```
    return 0;
```

```
}
```

Q2. Write a program to count the number of alphabets in a file.

Sol2:

```
#include <iostream>
```

```
#include <fstream>
```

```
#include <cctype> //for isalpha()
```

```
using namespace std;
```

```
//User-defined function
```

```

void countAlphabets() {
    ifstream file("NOTES.TXT");

    if (!file) {
        cout << "Error opening file!" << endl; return;
    }

    char ch;
    int count = 0;

    // Read file character by character while
    (file.get(ch)) {
        if (isalpha(ch)) {
            count++;
        }
    }

    file.close();

    cout << "Number of alphabets in file: " << count << endl;
}

int main() {
    countAlphabets();

    return 0;
}

```

Q3. Write a program to copy the contents of one file to another.

Sol 3:

```
#include <iostream>
```

```
#include <fstream> using
```

```
namespace std;
```

```
int main() {
```

```
    ifstream source("source.txt"); // file to read from
```

```
    ofstream destination("dest.txt"); // file to write into
```

```
    if (!source) {
```

```
        cout << "Error opening source file!" << endl; return 1;
```

```
    }
```

```
    if (!destination) {
```

```
        cout << "Error opening destination file!" << endl;
```

```
        return 1;
```

```
    }
```

```
    char ch;
```

```
    //Copy character by character while
```

```
    (source.get(ch)) {
```

```
        destination.put(ch);
```

```
    }
```

```
    cout << "File copied successfully!" << endl;
```

```
source.close();  
destination.close();  
  
return 0;  
}
```

Q4. Write a program to implement I/O operations on characters. Input a string, calculate its length, store it in a file, and fetch the stored characters from it.

Sol 4:

```
#include <iostream>  
  
#include <fstream>  
  
#include <cstring> using  
namespace std;  
  
int main() { char  
    str[100];  
  
    cout << "Enter a string: ";  
    cin.getline(str, 100);  
  
    int len = strlen(str);  
    cout << "Length = " << len << endl;  
  
    ofstream fout("string.txt");  
    fout << str; fout.close();  
  
    ifstream fin("string.txt"); char
```

```
ch;

cout << "File content: ";

while(fin.get(ch)) {
    cout << ch;
}

fin.close();
return 0;
}
```

Q5. Write a program to perform file pointer operations using seekg(), seekp(), and tellg().

Sol 5:

```
#include <iostream>

#include <fstream> using
namespace std;

int main() {

    // (a) Create file with A-Z and read 10th character of stream
    fout("file.txt");

    for(char ch = 'A'; ch <= 'Z'; ch++) { fout <<
        ch;
    }
```

```
fout.close();
```

```
ifstream fin("file.txt");
```

```
fin.seekg(9,ios::beg); //10th character (index starts from 0) char ch;
```

```
fin.get(ch);
```

```
cout << "(a) 10th character: " << ch << endl; fin.close();
```

```
// (b) Overwrite 5th character using seekp()
```

```
fstream file("file.txt", ios::in | ios::out);
```

```
file.seekp(4, ios::beg); // 5th position
```

```
file.put('X');      // overwrite with 'X'
```

```
file.close();
```

```
// Display updated content
```

```
fin.open("file.txt");
```

```
cout << "(b) Updated content: ";
```

```
while(fin.get(ch)) cout << ch;
```

```
cout << endl;
```

```
fin.close();
```

```
// (c) Find file size using tellg()
```

```
fin.open("file.txt");
```

```
fin.seekg(0, ios::end); // go to end
```

```
cout << "(c) File size: " << fin.tellg() << " bytes" << endl;
```

```
fin.close();
```

```
//(d) Read last character using seekg()
```

```
fin.open("file.txt");  
fin.seekg(-1, ios::end); // last character  
fin.get(ch);  
cout << "(d) Last character: " << ch << endl; fin.close();
```

```
// (e) Create data.txt with multiple lines  
ofstream data("data.txt");  
data << "Hello World\n"; data  
<< "This is a test file\n";  
data << "C++ File Handling Example\n";  
data.close();
```

```
ifstream dfile("data.txt");
```

```
// (i) Move pointer to 10th byte dfile.seekg(9,  
ios::beg);
```

```
// (ii) Display current position  
cout << "(e-ii) Current position: " << dfile.tellg() << endl;
```

```
// (iii) Read remaining content  
cout << "(e-iii) Remaining content:\n";  
while(dfile.get(ch)) { cout  
    << ch;  
}
```

```
dfile.close();
```

```
    return 0;
}
```

Q6. Write a program to demonstrate tellp() and seekp() in file handling.

Sol6:

```
#include <iostream>
```

```
#include <fstream> using
namespace std;
```

```
int main() {
    fstream file("hello.txt", ios::out);

    string text = "Hello World";

    for(int i = 0; i < text.length(); i++) {
        file.put(text[i]);

        cout << "Position: " << file.tellp() << endl;
    }
}
```

```
file.close();
```

```
// Replace "World" with "C++"
```

```
fstream file2("hello.txt", ios::in | ios::out);
```

```
file2.seekp(6, ios::beg); // start of "World" file2
```

```
<< "C++ ";
```



```
file2.close();
```

```
cout<<"Replacement done!";
```

```
return 0;
```

```
}
```